

The Orbital Propellant Kelp Farm

A Solar-Powered Water-Electrolysis Depot for In-Space Refuelling

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Classification: Speculative engineering, constrained to known physics

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Abstract

A rocket leaving Earth spends most of its mass carrying the fuel it needs for the second half of its journey — fuel that must itself be lifted, at ruinous cost, out of the deepest gravity well humans routinely climb. The Orbital Propellant Kelp Farm severs that penalty by manufacturing propellant where it is used. The depot receives water — the easiest substance to mine from asteroids and the lunar poles, and the safest to store — and splits it with electricity from vast solar arrays into hydrogen and oxygen, the highest-performance chemical propellant known. The gases are liquefied and banked in sun-shielded cryogenic tanks, and departing spacecraft top up in orbit rather than hauling a full load off the ground. We describe the electrolysis and cryogenic architecture, the 'kelp frond' solar canopy that names the station, the boil-off management that is the depot's central engineering battle, and a staged path from a water-launched demonstrator to an ice-fed orbital economy. Every subsystem has flown; the depot is an integration and logistics problem, not a physics one — which is why it may be the first genuinely industrial machine humanity builds off Earth.

1. Introduction: The Gas Station Comes First

Frontiers are not opened by explorers but by supply. The American West was won by water tanks and coaling stations along the railroad; the age of motoring waited on the filling station as much as on the engine. Spaceflight has reached the same threshold. We can build the ships; what we lack is the gas station in the sky, and its absence quietly dominates the economics of every deep-space mission, because a vehicle that must carry all its propellant from the surface is mostly propellant by mass, leaving little for cargo.

The fix has been understood for decades and is captured in a single phrase: in-space propellant depots. Refuel in orbit, and a spacecraft need only reach orbit half-empty, then fill up for the real journey — to the Moon, to Mars, to the asteroids — with propellant that never had to be lifted through the atmosphere at all, if it was made from ice already up there. NASA and industry studies have repeatedly found depots to be enabling for beyond-Earth exploration, and the required technologies have been maturing steadily under long-running cryogenic-fluid-management programmes (Chato, 2008; Notardonato et al., 2017). The Kelp Farm is what those studies describe when drawn as a working factory.

2. Concept Description

The PD-1 'Poseidon' is a solar-electric refinery station in a stable low or medium Earth orbit. Its defining feature, and its nickname, is the canopy: hectares of thin-film photovoltaic 'fronds' spread on a lightweight truss, swaying to track the Sun and drinking the raw energy that powers everything else. Beneath the canopy sit three functional zones. The intake dock receives water delivered as dense, benign cargo — from Earth at first, from mined ice later — and purifies it. The electrolysis hall splits that water

into hydrogen and oxygen. The cryogenic tank farm liquefies and stores the gases, shielded from the Sun, until a customer arrives to fill up at the transfer boom.

The elegance of choosing water as the feedstock is that it is the friendliest molecule in space logistics: inert, non-explosive, radiation-shielding, drinkable, and — critically — the most abundant volatile at the asteroids and lunar poles that will eventually feed the depot. A station that traffics in water can serve as fuel refinery, radiation shelter, and life-support reservoir at once, three businesses sharing one tank.

3. The Chemistry and the Cold

Electrolysis is old, robust, and already spaceflight-proven: the International Space Station has generated its crew's oxygen by splitting water for years, so the core reaction — water plus electricity yields hydrogen and oxygen — needs no invention, only scaling to industrial throughput (Chato, 2008). The oxygen becomes oxidiser; the hydrogen becomes the highest-specific-impulse chemical fuel there is. A depot's output is therefore limited chiefly by its electrical power, which is why the solar canopy is so large: propellant, here, is simply stored sunlight in chemical form.

The true adversary is not chemistry but heat. Liquid hydrogen must be kept below about 20 kelvin and liquid oxygen below about 90 kelvin, and in sunlight, tanks relentlessly absorb warmth and their contents boil away — the 'boil-off' that has defeated naive depot proposals for half a century. The Kelp Farm answers with the same toolkit NASA has been flight-qualifying: multilayer insulation, sunshields that keep tanks in permanent shadow, vapour-cooled shielding, and above all active refrigeration. Cryocoolers powered by the canopy intercept incoming heat and reliquefy vapour, achieving the 'zero-boil-off' condition demonstrated in ground testbeds and now maturing toward flight (Notardonato et al., 2017; Mustafi et al., 2010). Zero-boil-off storage is the hinge on which the entire concept turns, and it is precisely the technology that has advanced most in the last decade.

Transferring cryogenic propellant between vehicles in weightlessness poses its own subtle problem: without gravity, liquid and vapour will not obligingly separate, so the depot uses settling thrusts, surface-tension propellant-management devices, and careful thermal conditioning to deliver liquid — not froth — to a customer's tanks. In-space cryogenic transfer is the subject of active flight demonstrations, and its maturation is what turns a storage tank into a functioning pump (Notardonato et al., 2017).

4. The Machine Itself

Follow a molecule of water through the station. It arrives in a delivery tanker and is pumped into a holding bladder, filtered of the grit and salts that mined ice will carry. Drawn into the electrolysis hall, it meets a stack of cells energised by the canopy; hydrogen bubbles from one electrode, oxygen from the other, and the two gases part company for good. Each stream flows to its liquefier, where staged cryocoolers chill it past its boiling point into the tank farm — hydrogen in the deepest-shielded, coldest tanks, oxygen in the merely frigid ones. There the propellant waits, its slow warming continuously undone by refrigeration, until a spacecraft docks at the transfer boom and the depot pumps it a full load in hours. The station's rhythm is that of any refinery: intake, convert, store, dispense — only the sunlight is free and the gravity absent.

5. Operations Concept

Phase one launches water and proves the machine: a modest depot in low Earth orbit, fed by water lofted as ordinary cargo, demonstrating electrolysis, zero-boil-off storage, and cryogenic transfer to a customer vehicle. Even this version pays, because water is dense, cheap, and safe to launch, and the depot converts it to propellant on orbit. Phase two switches feedstock: as lunar-polar and asteroid ice mining come online, the depot begins receiving ice-derived water that never climbed out of Earth's gravity well, collapsing the cost of propellant in orbit. Phase three replicates the station along the traffic lanes — a depot in high orbit, another at a Lagrange point, a lunar-orbit outpost — until refuelling in space is as unremarkable as it is on any highway.

Parameter	Phase 1 (LEO demo)	Phase 3 (Ice-fed depot)
Feedstock	water launched from Earth	mined lunar/asteroid ice
Power (solar canopy)	~100 kW	megawatts
Products	LH2 + LO2	LH2 + LO2 (+ water resale)
Storage	zero-boil-off, active cooled	zero-boil-off, scaled
Transfer	to one docked vehicle	multi-berth fuelling
Secondary role	—	radiation shelter / water depot

Table 1. Representative parameters for the staged build-out.

6. Honest Difficulties

Three problems deserve candour. Boil-off is the perennial one: zero-boil-off storage works in ground testbeds and is maturing in flight, but keeping liquid hydrogen cold for months in orbit at industrial scale is unforgiving, and every watt of cooling must come from the canopy (Mustafi et al., 2010). Cryogenic transfer in microgravity — settling the liquid, avoiding vapour locks, conditioning warm tanks without wasting propellant — is genuinely hard and only now being demonstrated in flight (Notardonato et al., 2017). And the depot's economics depend on a supply chain that does not yet exist: without cheap water delivery, and eventually cheap mined ice, the station is a solution waiting for its customers. The honest status is that the depot's physics is settled and its enabling technologies are converging fast, but its business case is chicken-and-egg with the mining industry it will one day feed on.

7. Pathway to Reality

This concept sits closer to reality than almost any other in the catalogue. Water electrolysis flies on the space station today; zero-boil-off and in-space cryogenic transfer are the explicit subjects of active NASA technology programmes and commercial flight demonstrations; orbital refuelling is already central to at least one operational heavy-launch architecture (Notardonato et al., 2017; Chato, 2008). A water-fed demonstrator depot is plausibly a this-decade or early-next-decade project; an ice-fed depot follows the maturation of lunar and asteroid mining. Of all the machines in this collection, the Kelp Farm is the one most likely to be built for profit rather than prestige — the surest sign that a technology has truly arrived.

8. Conclusion

Great migrations follow the wells. The Orbital Propellant Kelp Farm is a bet that the second space age turns industrial the moment it stops carrying its own fuel and starts buying it on the way — that the

decisive machine is not a faster rocket but a tank of cold hydrogen, made from sunlight and water, waiting in orbit for anyone bound outward. When ships stop launching full and start launching light, topping up in the sky, the gas station will have done what gas stations always do: turn a frontier into a place people simply go.

References

Harvard referencing style (Cite Them Right).

- Chato, D.J. (2008) 'Cryogenic technology development for exploration missions', in *46th AIAA Aerospace Sciences Meeting and Exhibit*, AIAA 2008-951. Reno, NV: American Institute of Aeronautics and Astronautics.
- Mustafi, S., Canavan, E., Johnson, W., Kutter, B. and Shull, J. (2010) 'Subcooling cryogenic propellants for long duration space exploration', in *AIAA SPACE 2010 Conference and Exposition*, AIAA 2010-8869. Anaheim, CA: American Institute of Aeronautics and Astronautics.
- Notardonato, W.U., Swanger, A.M., Fesmire, J.E., Jumper, K.M., Johnson, W.L. and Tomsik, T.M. (2017) 'Zero boil-off methods for large-scale liquid hydrogen tanks using integrated refrigeration and storage', *IOP Conference Series: Materials Science and Engineering*, 278, 012012.